

Jason Biatek

Contact Information:

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Research software engineer specializing in formal methods, programming languages, language engineering, model-driven development, and safety-critical software systems. Over 10 years of experience designing compilers, DSLs, translators, verification tools, and analysis frameworks for aerospace, defense, and high-assurance software. Deep expertise translating between modeling languages, building developer tooling, and applying theorem proving and model checking to complex engineering systems.

SOFTWARE ENGINEERING PROJECTS

SysMLv2 model representation and Lustre translation tool (Haskell)

- Haskell tool for parsing, manipulating, and translating to and from SysMLv2
- Ownership of the parser, extensive work on AST, interesting challenges between SysML's object-oriented nature and Haskell's pure functional programming semantics
- I planned and executed a months-long project to translate a Lustre model to SysMLv2, with full fidelity of Lustre semantics, including static analysis to correctly order computations

Domain-specific language for XSAM output (Kotlin, web microservice)

- Developed a Kotlin DSL for expressing data in an XML format, making it easy for developers to write templates that were type-safe and well-formatted
- The microservice provided file conversions, primarily taking Excel spreadsheets and converting them to XML for import into another tool
- Small web frontend written in Vue for manual file upload
- I joined the project very early and helped write the templating code as well as many of the data conversions themselves

Text2Test (C#, Antlr, Java)

- A structured natural language for writing software requirements, which reads like English to humans but can be parsed and analyzed by software
- Also included work on an Eclipse Xtext plugin providing an editor for this language
- I worked with the large existing code base, made suggestions on how to make the requirement language clearer, and added new grammar and features to the analysis tool and the plugin

TPlex research tool (Java)

- Parses PLEXIL code (an autonomous planning language developed by NASA), transforms and optimizes in intermediate format, then translates to Java and Lustre
- Searches for test inputs that satisfy properties using JKind
- Designed to be flexible and extendable for research purposes (since we don't know what the right answers are)
- I designed and coded nearly the entire tool, with advice from professors

AREAS OF EXPERIENCE

Specialties: Data & code translation, language engineering, domain-specific languages, testing, model-driven development, model checking, formal methods, safety-critical systems

Programming Languages: Kotlin, Java, Haskell, Python, C#, Swift, Lustre, Powershell, Bash, Standard ML, C, OCaml, Prolog, PHP, Python, Ruby, Perl, MySQL, Lisp / Scheme, anything really

Engineering Tools: VS Code, Git, JetBrains MPS & IntelliJ, Eclipse, Visual Studio, SVN

Specification Languages: SysMLv2, Rhapsody, Simulink, StateFlow

Model Analysis Tools: HiLiTE, JKind, Z3, Java PathFinder, NuSMV, KLEE, Reactis

EDUCATION

Master of Computer Science, 2013, University of Minnesota, Twin Cities

B.A. Computer Science (Music minor), with distinction, 2010, University of Minnesota, Morris

EMPLOYMENT

Research Engineer, Galois

August 2025—July 2026

Senior Software Engineer, Itemis Inc.

August 2022—June 2025

Research Scientist, Honeywell

February 2017—July 2022

Research & Teaching Assistant, University of Minnesota

Jan 2011—Jan 2017

Summer Intern, NASA Ames

June—August 2013

Summer Intern, Medtronic

May—August 2012

Research Assistant, University of Minnesota Morris

May 2008—May 2010

EMPLOYMENT DETAILS

Research Engineer, Galois, August 2025—July 2026

At Galois, I contributed to multiple projects, including a Haskell tool for working with SysMLv2 where I assisted with parsing and translation work for textual SysML, as well as designing and implementing a Lustre to SysMLv2 translation. I also worked on a project that created a detailed mathematical network model and used formal methods to check properties and do analysis, where I was in charge of creating a REST API for accessing the tool through an easy-to-use interface, as well as implementing features like a symbolic comparison between network model snapshots and reachability analysis between network nodes.

Senior Software Engineer, Itemis Inc., August 2022—June 2025

As an employee at a very small company (just 3 employees in the US!) I wore many hats in addition to software development, including customer support, outreach at conventions, and generally lending a hand where needed. Some projects I worked on included a Kotlin microservice backend, advanced TypeScript/JavaScript tooling to smooth over a custom backend's awkward data model and make it feel more native, some Vue work on web frontends, writing scripts for customers to do specific data import into a threat and risk assessment tool, specifying an XML format with a formal schema, and becoming the "local expert" in an Eclipse-based traceability tool.

Research Scientist, Honeywell, February 2017—July 2022

Honeywell has developed a language called CLEAR for writing software requirements, as well a tool called Text2Test for analysis of requirements written in that language. CLEAR reads like a natural language specification, but is specified by a grammar and semantics which make it possible for software to parse and understand. Text2Test was built on top of HiLiTE, an internal tool at Honeywell used for requirements-based verification written in C#. It takes CLEAR requirements and translates them into the existing structures that HiLiTE uses, which enables analysis and automatic test generation.

I was initially hired as an intern to work on this tool, and was hired full-time afterward. I worked on finding and fixing bugs in Text2Test, as well as adding new features. Typically, this included modifications to the grammar for parsing CLEAR and code to implement the semantics of the new grammar. I needed to quickly get up to speed on a large existing codebase, and work within that codebase to add new features as well as find and fix bugs that came up during development. I also worked on a text editor for CLEAR, implemented as an Eclipse plug-in built with Xtext. In addition to this more direct coding work, I also was involved in the bigger picture of finishing the language specification of CLEAR, putting forward my thoughts on how best to make the language unambiguous and easy to use.

Research Assistant, University of Minnesota, Jan 2011—Jan 2017

I worked on automated test case generation for PLEXIL, a planning language created by NASA. Automatic test case generation using SMT solvers is a well-known technique for automatically creating test inputs that satisfy a given property, but PLEXIL plans are intended for use in an autonomous system. As such, PLEXIL plans often require long test cases to achieve code coverage, because the system needs to make sure that both itself and the environment are in an expected state before proceeding. Since SMT solvers require exponentially more time as the length of a test case increases, the straightforward approach is not sufficient.

I wrote a translator to convert PLEXIL plans into Lustre, which is a modeling language used by the model checker JKind. With this, we could automatically generate tests for PLEXIL plans, but scalability quickly became a limiting factor. I worked on using incremental test case generation, working toward an overall goal one achievable segment at a time. With this technique, it is critical that each segment makes progress toward the goal. I hoped to show that knowledge of the plan and the semantics of the language could be used to guide the search for a particular test case.

EMPLOYMENT DETAILS

Research Assistant, University of Minnesota Morris, May 2008—May 2010

I worked with Dr. Kristin Lamberty on DigiQuilt, an educational software application intended to help elementary school students learn symmetry and fractions while constructing quilt patterns. I fixed the incomplete Java port of the software and added many new features, including a large display that we tested with a group of students at a local elementary school. This work was the subject of a “work-in-progress” paper at CHI 2010 and a paper at IDC 2011.

PUBLICATIONS

Analysis and testing of PLEXIL plans, Jason Biatek, Michael W. Whalen, Mats P. E. Heimdahl, Sanjai Rayadurgam, Michael R. Lowry, FormaliSE 2014: Proceedings of the 2nd FME Workshop on Formal Methods in Software Engineering, June 2014, <https://dl.acm.org/doi/abs/10.1145/2593489.2593496>

Requirements-Driven Model Checking and Test Generation for Comprehensive Verification, Devesh Bhatt, Hao Ren, Anitha Murugesan, Jason Biatek, Srivatsan Varadarajan & Natarajan Shankar, NASA Formal Methods 2022, May 2022, https://link.springer.com/chapter/10.1007/978-3-031-06773-0_31

DesCert: Design for Certification, Natarajan Shankar, Devesh Bhatt, Michael Ernst, Minyoung Kim, Srivatsan Varadarajan, Suzanne Millstein, Jorge Navas, Jason Biatek, Huascar Sanchez, Anitha Murugesan, Hao Ren, CSL Technical Report SRI-CSL-2022-01, March 2022, <https://arxiv.org/abs/2203.15178>

Helping system engineers bridge the peaks, Neha Rungta, Oksana Tkachuk, Suzette Person, Jason Biatek, Michael W. Whalen, Joseph Castle, Karen Gundy-Burlet, TwinPeaks 2014: Proceedings of the 4th International Workshop on Twin Peaks of Requirements and Architecture, June 2014, <https://dl.acm.org/doi/abs/10.1145/2593861.2593863>

Integrating Statechart Components in Polyglot, Daniel Balasubramanian, Corina S. Păsăreanu, Jason Biatek, Thomas Pressburger, Gabor Karsai, Michael Lowry & Michael W. Whalen, NASA Formal Methods 2012, 2012, https://link.springer.com/chapter/10.1007/978-3-642-28891-3_26

Using a large display in the periphery to support children learning through design, K. K. Lamberty, Stephen Adams, Jason Biatek, Katherine Froiland, Jay Lapham, IDC '11: Proceedings of the 10th International Conference on Interaction Design and Children, June 2011, <https://dl.acm.org/doi/abs/10.1145/1999030.1999038>

Encouraging awareness of peers' learning activities using large displays in the periphery, K. K. Lamberty, Katherine Froiland, Jason Biatek, Stephen Adams, CHI EA '10: CHI '10 Extended Abstracts on Human Factors in Computing Systems, April 2010, <https://dl.acm.org/doi/abs/10.1145/1753846.1754034>